Brigham and Women's Hospital
Founding Member, Mass General Brigham

Psychiatry Neuroimaging Laboratory

HARVARD MEDICAL SCHOOL

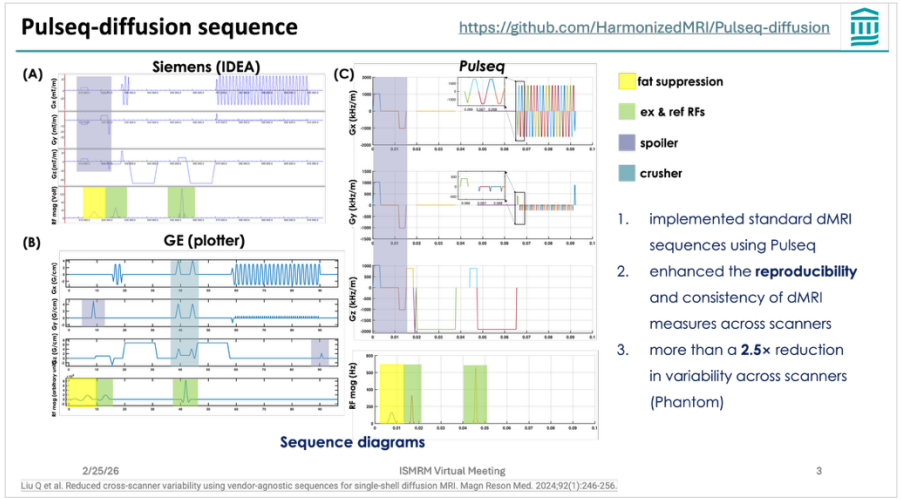
Pulseq for High-Resolution Diffusion Acquisitions

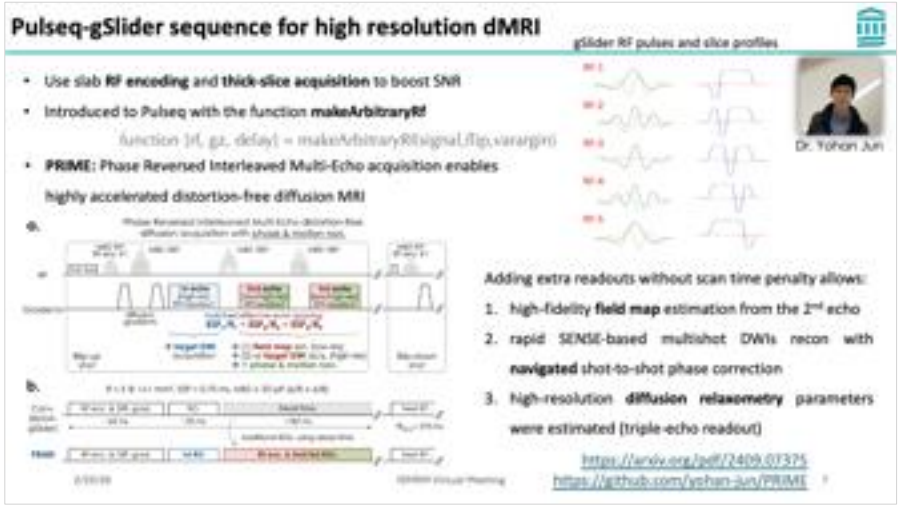
Qiang Liu

2026/02/26

Brigham and Women's Hospital, Harvard Medical School

Outline


Pulseq-diffusion sequence
https://github.com/HarmonizedMRI/Pulseq-diffusion
(A)
Siemens (IDEA)
(B)
GE (plotter)
(C)
Pulseq
fat suppression
ex & ref RFs
spoiler
crusher
1. implemented standard dMRI sequences using Pulseq
2. enhanced the reproducibility and consistency of dMRI measures across scanners
3. more than a 2.5× reduction in variability across scanners (Phantom)
Sequence diagrams
2/25/26
ISMRM Virtual Meeting
3
Liu Q et al. Reduced cross-scanner variability using vendor-agnostic sequences for single-shell diffusion MRI. Magn Reson Med. 2024;92(1):246-256.


Pulseq-gSlider sequence for high resolution dMRI
gSlider RF pulses and slice profiles
Use slab RF encoding and thick-slice acquisition to boost SNR
Introduced to Pulseq with the function makeArbitraryRf
PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI
Dr. Yohan Jun
Adding extra readouts without scan time penalty allows:
1. high-fidelity field map estimation from the 2nd echo
2. rapid SENSE-based multishot DWIs recon with navigated shot-to-shot phase correction
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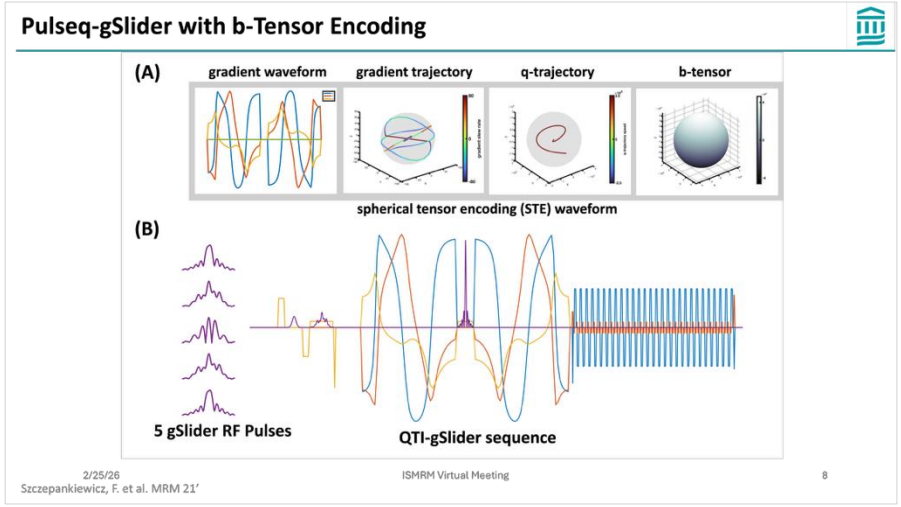
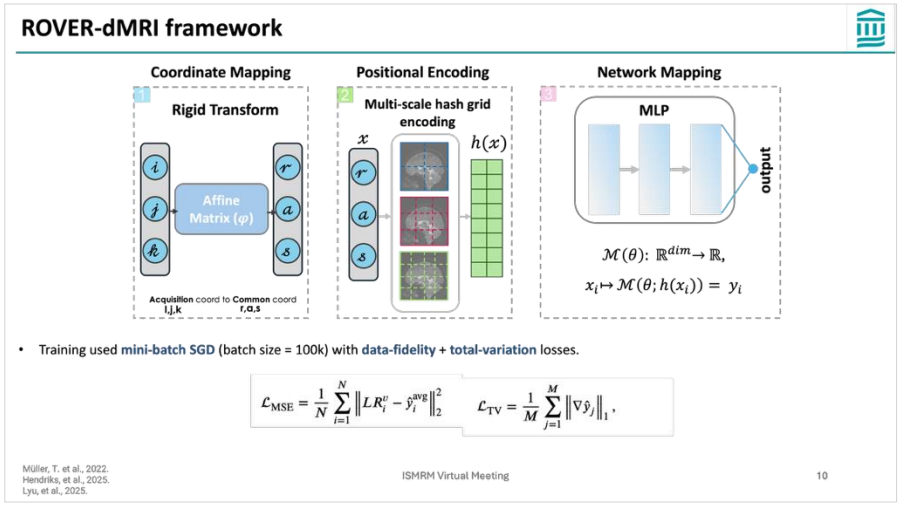

Pulseq-gSlider with b-Tensor Encoding
(A)
gradient waveform
gradient trajectory
q-trajectory
b-tensor
spherical tensor encoding (STE) waveform
(B)
5 gSlider RF Pulses
QTI-gSlider sequence
2/25/26
ISMRM Virtual Meeting
8
Szczepankiewicz, F. et al. MRM 21'


ROVER-dMRI framework
Coordinate Mapping
Positional Encoding
Network Mapping
Rigid Transform
Affine Matrix (φ)
Acquisition coord to Common coord
i,j,k
r,a,s
Multi-scale hash grid encoding
h(x)
MLP
output
$\mathcal{M}(\theta)$: $\mathbb{R}^{dim} \to \mathbb{R}$,
$x_i \mapsto \mathcal{M}(\theta; h(x_i)) = y_i$
Training used mini-batch SGD (batch size = 100k) with data-fidelity + total-variation losses.
$\mathcal{L}_{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left\|LR_i^{\nu} - \hat{y}_i^{avg}\right\|_2^2$
$\mathcal{L}_{TV} = \frac{1}{M}\sum_{j=1}^{M}\left\|\nabla\hat{y}_j\right\|_1$,
Müller, T. et al., 2022.
Hendriks, et al., 2025.
Lyu, et al., 2025.
ISMRM Virtual Meeting
10

Pulseq-diffusion sequence

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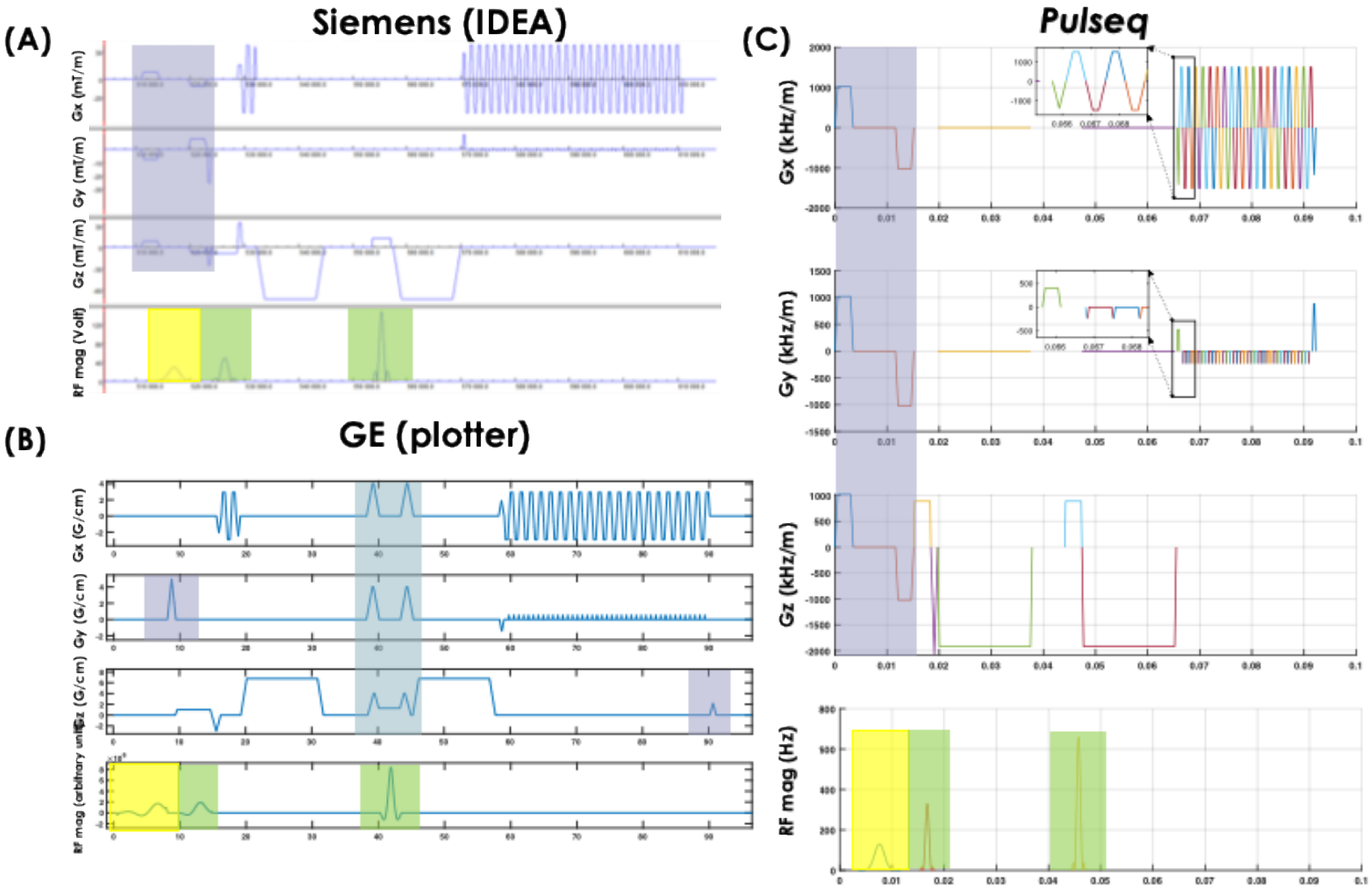

Sequence diagrams

fat suppression

ex & ref RFs

spoiler

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MRM paper

1. implemented standard dMRI sequences using Pulseq
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Pulseq-diffusion sequence

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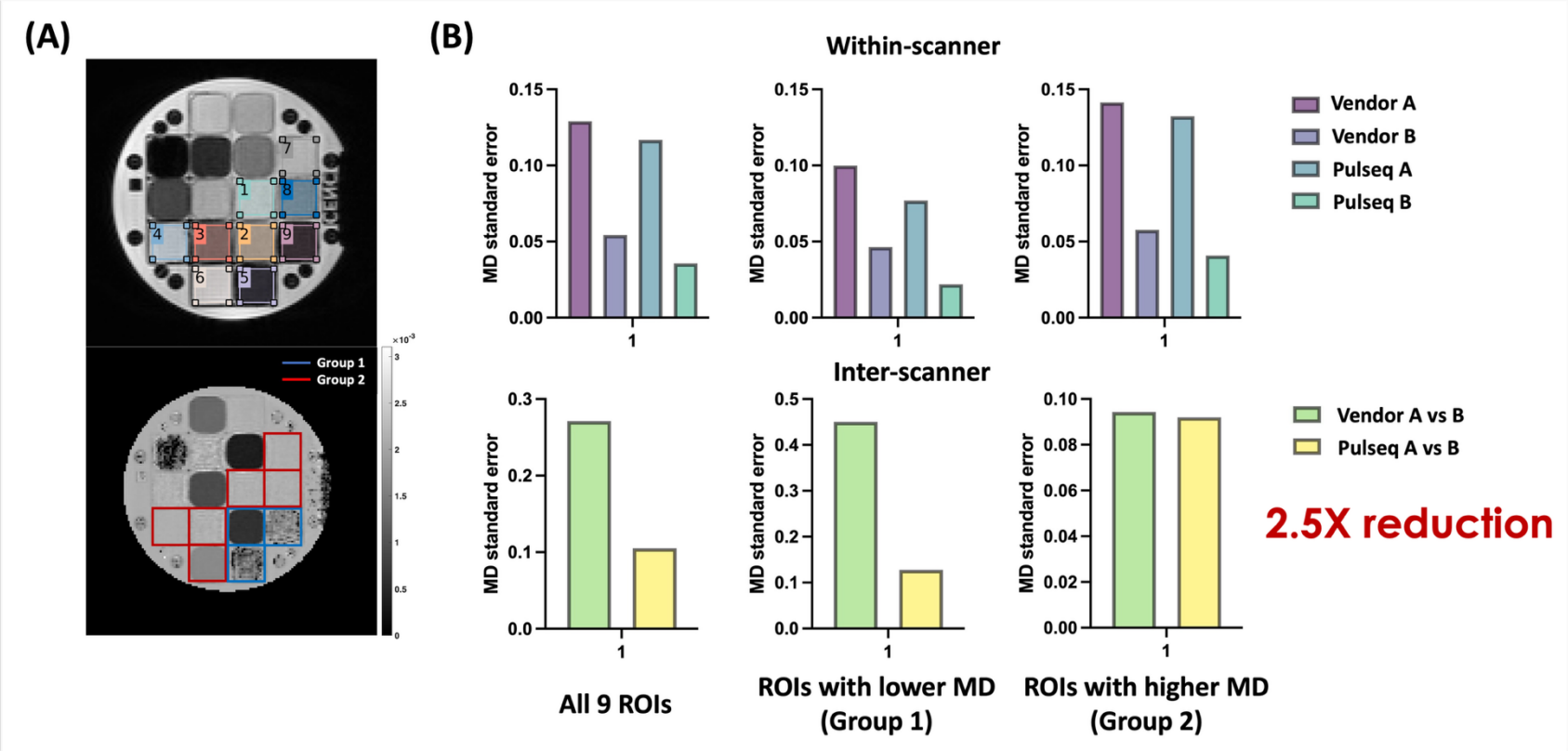
Phantom results



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Pulseq-diffusion sequence

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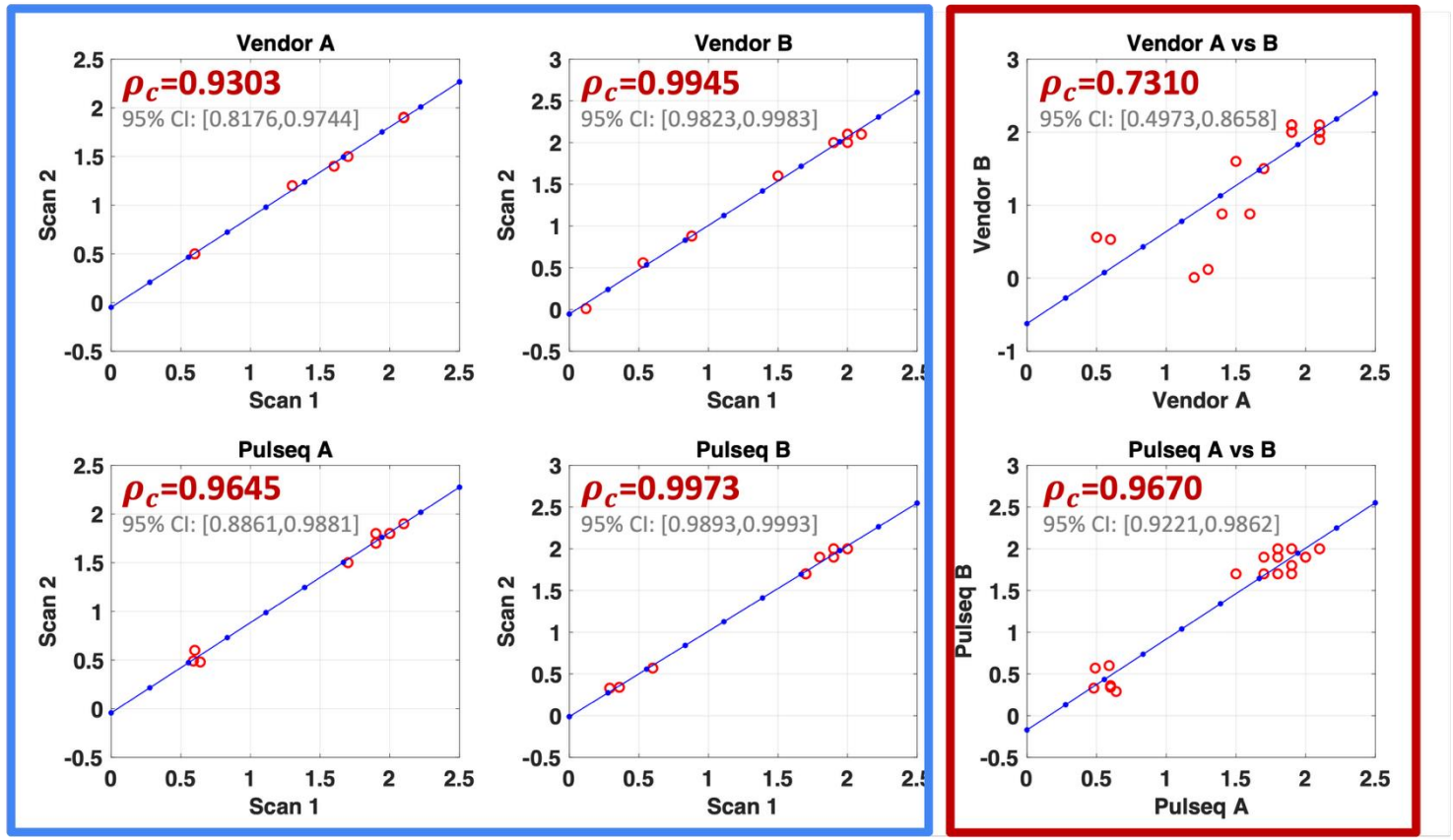
Phantom results

Lin's Concordance correlation (CCC)



Within-scanner **Inter-scanner**

- fat suppression
- ex & ref RFs
- spoiler
- crusher

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Pulseq-diffusion sequence

https://github.com/HarmonizedMRI/Pulseq-diffusion

Convert ***seq (Pulseq)*** to ***pge (GE)***

https://github.com/HarmonizedMRI/pge2

```
% create .seq file
fn = '/Users/Desktop/se_28sli_0p4mm_R3_22dir_2_shot_pa.seq';
seq = mr.Sequence();
read(seq,fn);

% scanner hardware settings
psd_rf_wait = 100e-6;  % RF-gradient delay, scanner specific (s)
psd_grd_wait = 100e-6; % ADC-gradient delay, scanner specific (s)
b1_max = 0.25;        % Gauss
g_max = 35;           % Gauss/cm
slew_max = 75;        % Gauss/cm/ms
coil = 'magnus';        % 'hrmbuhp' (UHP); 'xrm' (MR750)
sysGE = pge2.opts(psd_rf_wait, psd_grd_wait, b1_max, g_max, slew_max, coil);

% convert to Ceq object and check
ceq = seq2ceq(fn);
pge2.validate(ceq, sysGE, seq, [], 'plot', false);
% pge2.validate(ceq, sysGE);
% write to .pge file
pislquant = 100;  % number of ADC events for receive gain calibration
PNSweights = [1 1 1];  % direction weights
pge2.seq2ge(fn, sysGE, pislquant, PNSweights);
```

Key Parameters Across Scanners to Check

1. **Gradient Raster Time**

Siemens: multiple of **10 µs**

GE: multiple of **4 µs**

2. **RF Ringdown Time**

Pulseq default may differ from GE hardware requirement

3. **B0 field strength**

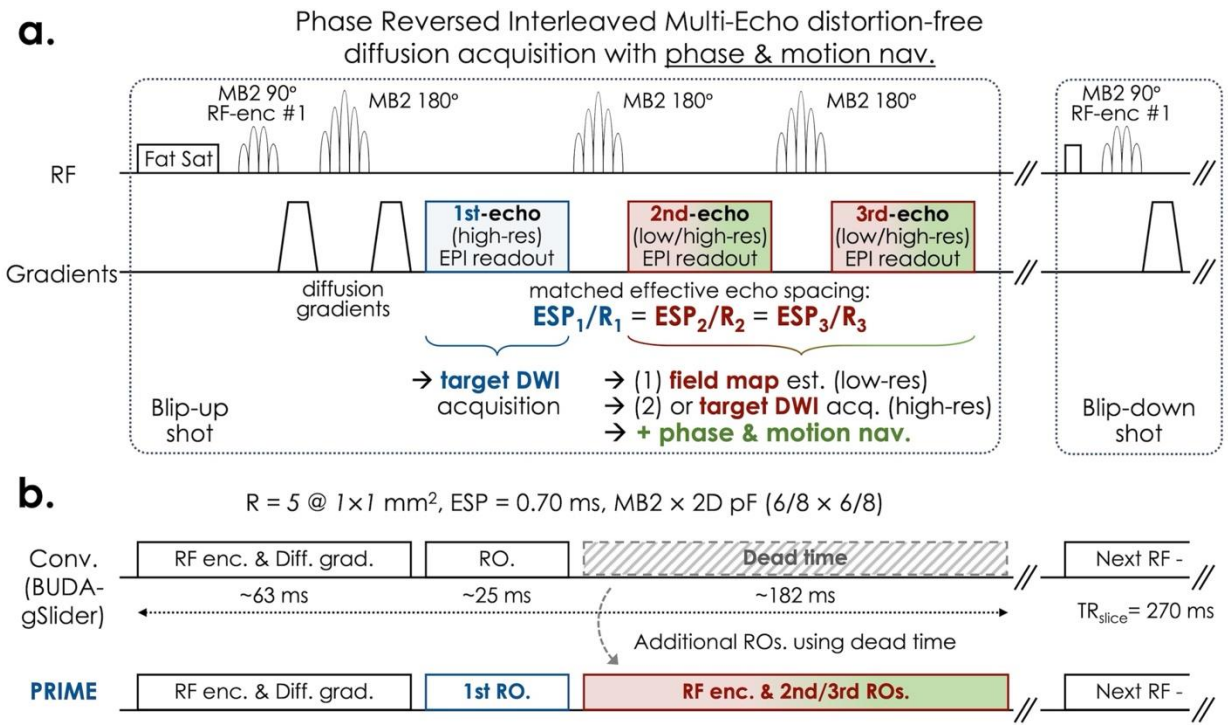
4. **ADC Dwell Time GE typical Nyquist dwell:** 2 µs, 4 µs …

Pulseq-gSlider sequence for high resolution dMRI

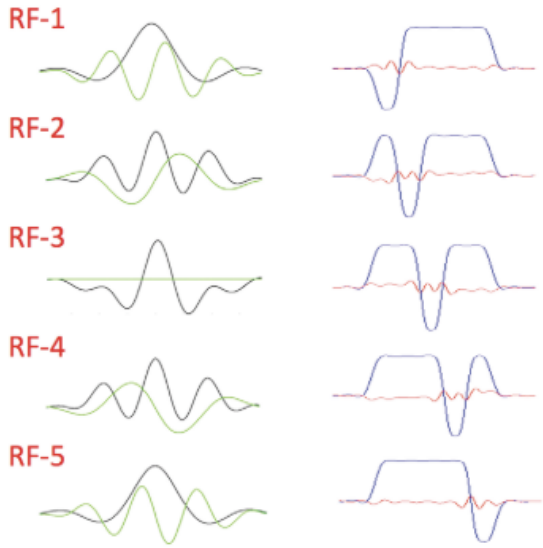
- Use slab **RF encoding** and **thick-slice acquisition** to boost SNR
- Introduced to Pulseq with the function **makeArbitraryRf**

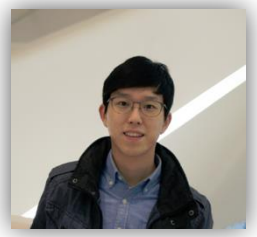
```
function [rf, gz, delay] = makeArbitraryRf(signal,flip,varargin)
```

- **PRIME:** Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI



gSlider RF pulses and slice profiles





Dr. Yohan Jun

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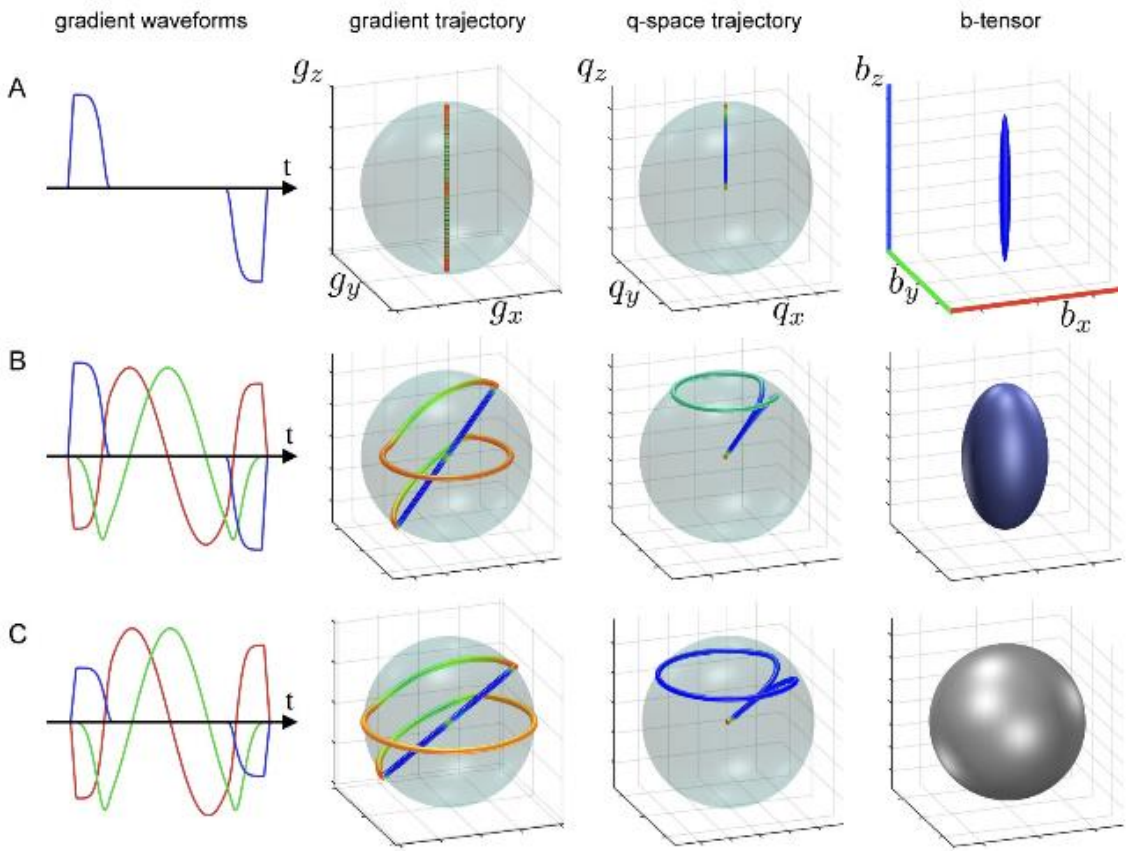
https://arxiv.org/pdf/2409.07375
https://github.com/yohan-jun/PRIME

Pulseq-gSlider with b-Tensor Encoding

- **Spherical Tensor Encoding (STE)** probes *microscopic diffusion anisotropy*
- Existing STE acquisitions suffer from **low spatial resolution**
- Key idea: STE encoding + gSlider slab RF encoding



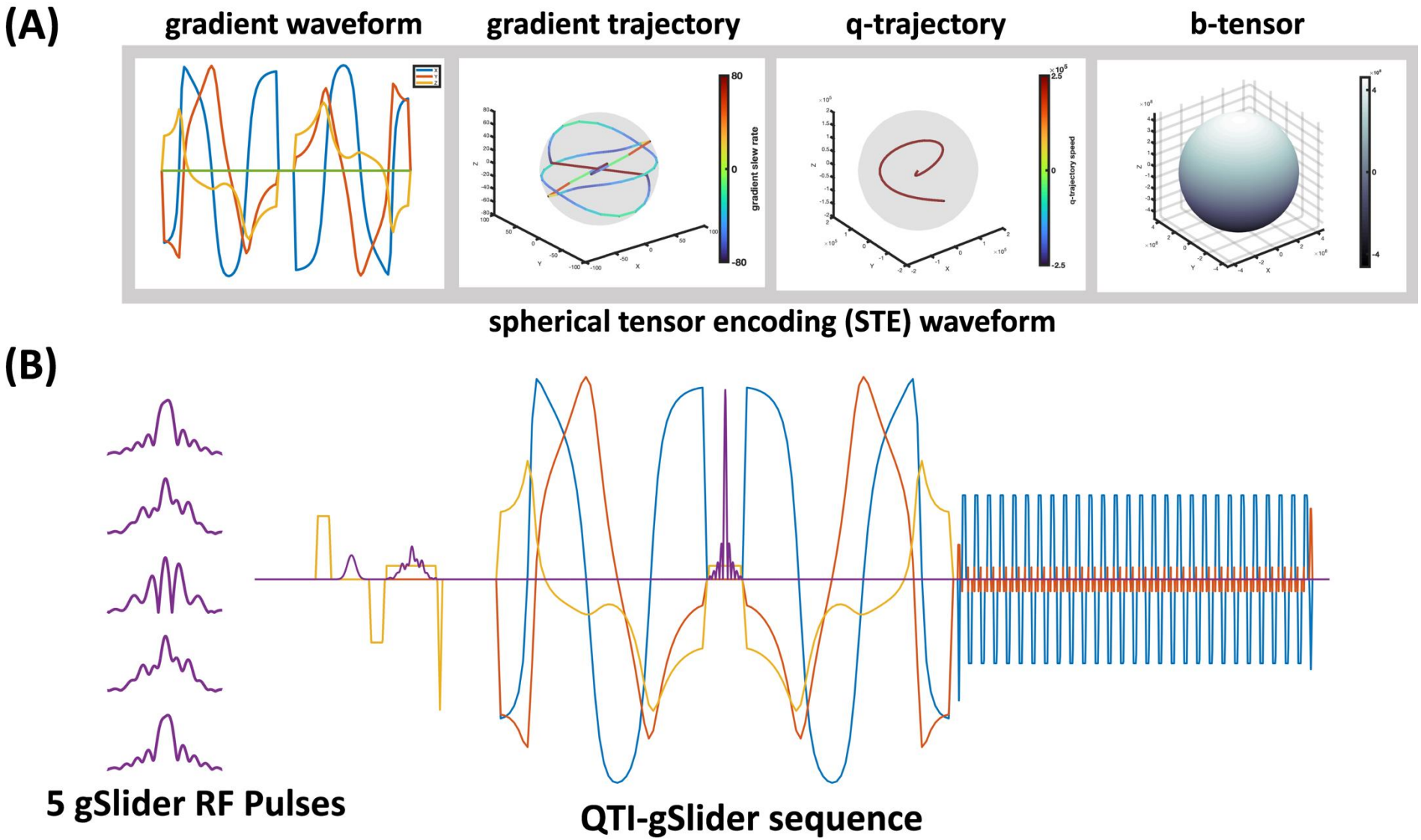
- **mr.makeArbitraryGrad** with STE waveforms using **NOW** toolbox

https://github.com/jsjol/NOW

```
for ds=1:dummy_count % ds=1: dummy scan, EPI ref; ds=2: b0 (AP); ds=3: b0 (PA); ds=4: QTI.

    gDiff_x_1=mr.makeArbitraryGrad('x', grad_diff_X(section1),'delay',0,'system',lims2);gDiff_x_1.first=0;gDiff_x_1.last=0;
    gDiff_x_2=mr.makeArbitraryGrad('x', grad_diff_X(section2),'delay',0,'system',lims2);gDiff_x_2.first=0;gDiff_x_2.last=0;

    gDiff_y_1=mr.makeArbitraryGrad('y', grad_diff_Y(section1),'delay',0,'system',lims2);gDiff_y_1.first=0;gDiff_y_1.last=0;
    gDiff_y_2=mr.makeArbitraryGrad('y', grad_diff_Y(section2),'delay',0,'system',lims2);gDiff_y_2.first=0;gDiff_y_2.last=0;


    gDiff_z_1=mr.makeArbitraryGrad('z', grad_diff_Z(section1),'delay',0,'system',lims2);gDiff_z_1.first=0;gDiff_z_1.last=0;
    gDiff_z_2=mr.makeArbitraryGrad('z', grad_diff_Z(section2),'delay',0,'system',lims2);gDiff_z_2.first=0;gDiff_z_2.last=0;
```


Westin et al., NIMG 16'
Szczepankiewicz, F. et al. MRM 21'

Pulseq-gSlider with b-Tensor Encoding



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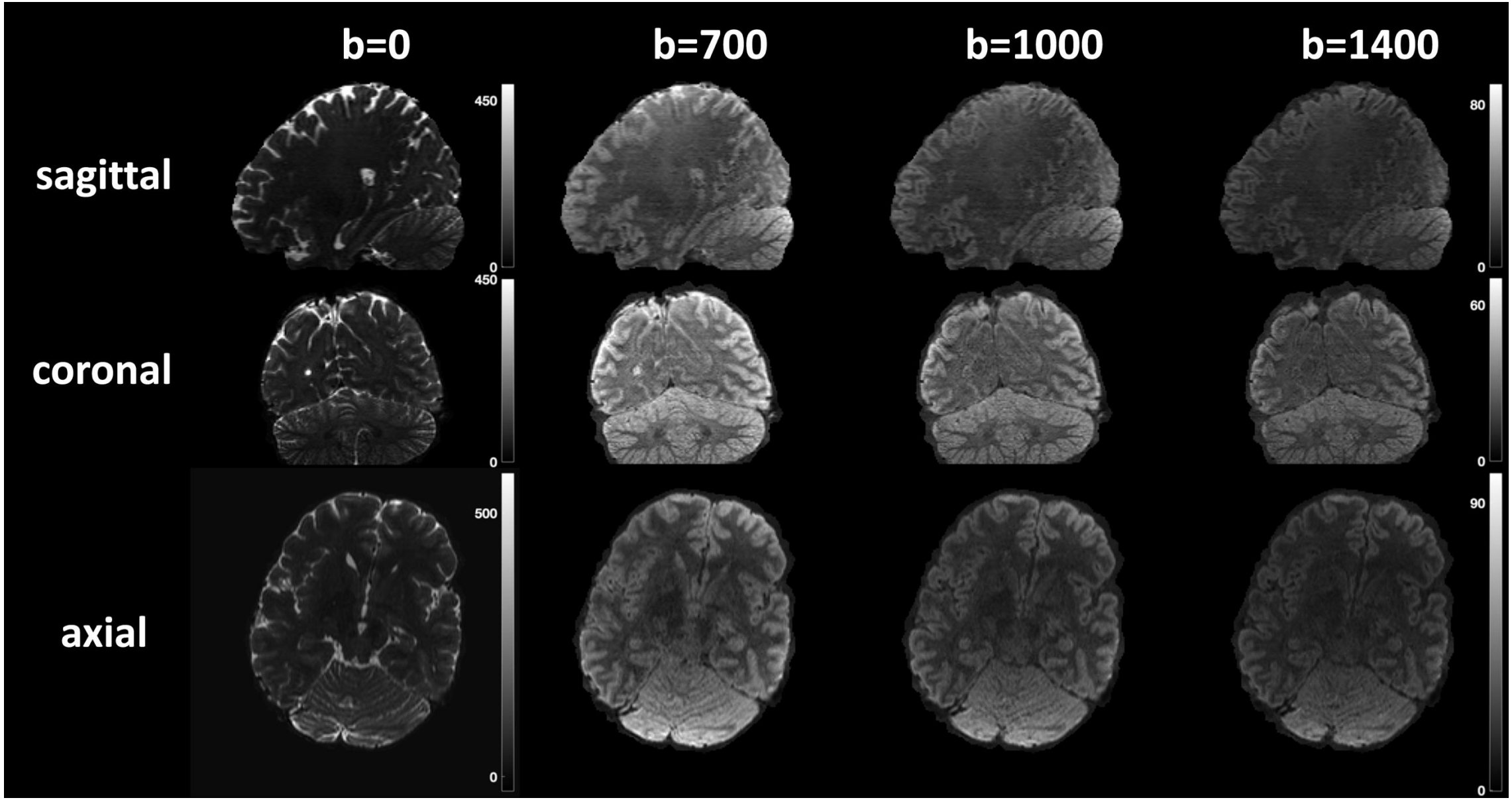
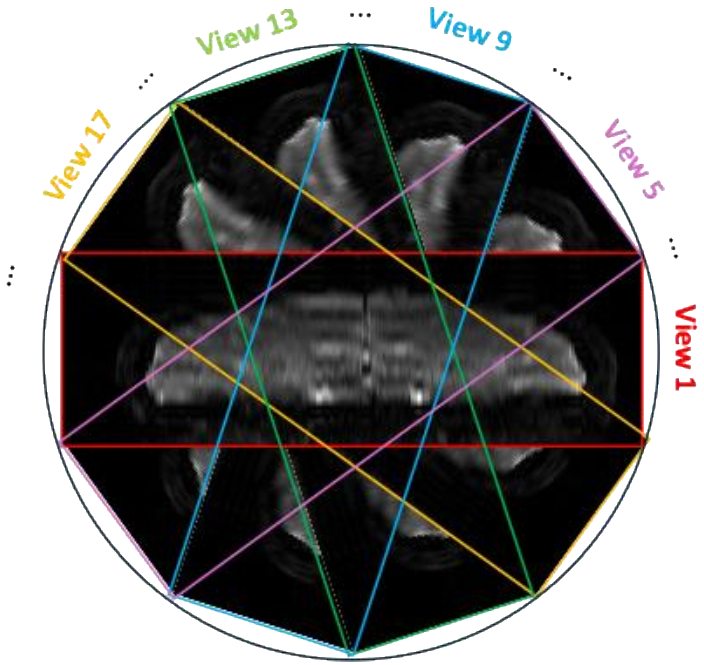


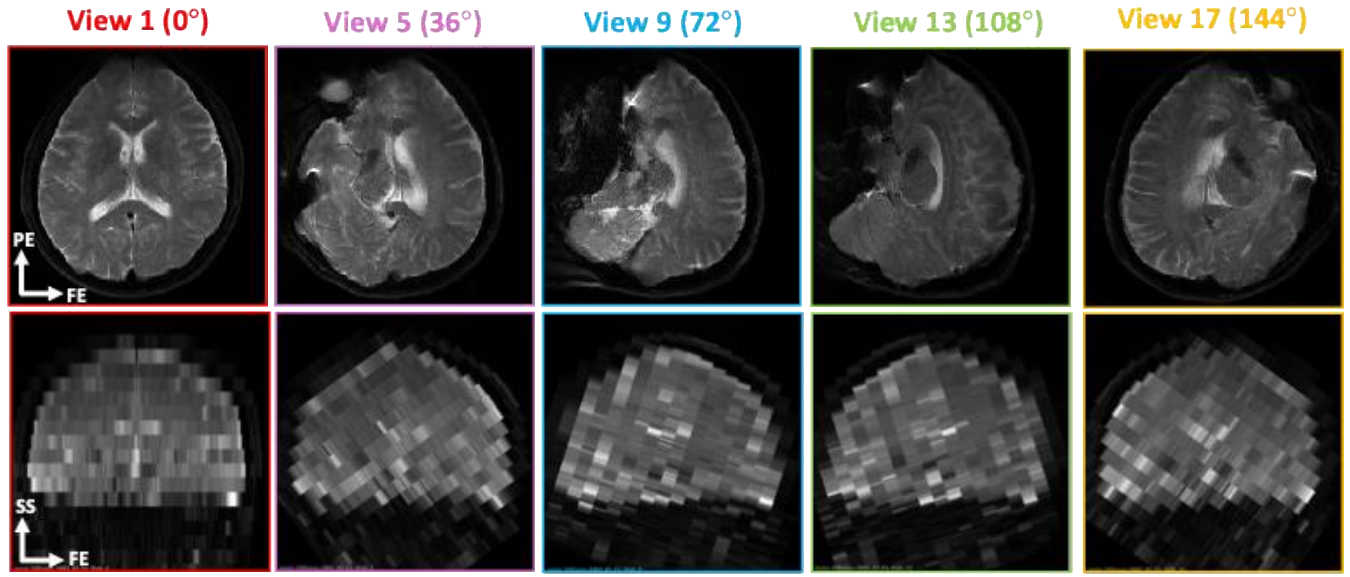
ROVER-dMRI: Rotating-View Super-Resolution Reconstruction for High-SNR Mesoscale Diffusion MRI

1. Target resolution: **hundreds of microns**

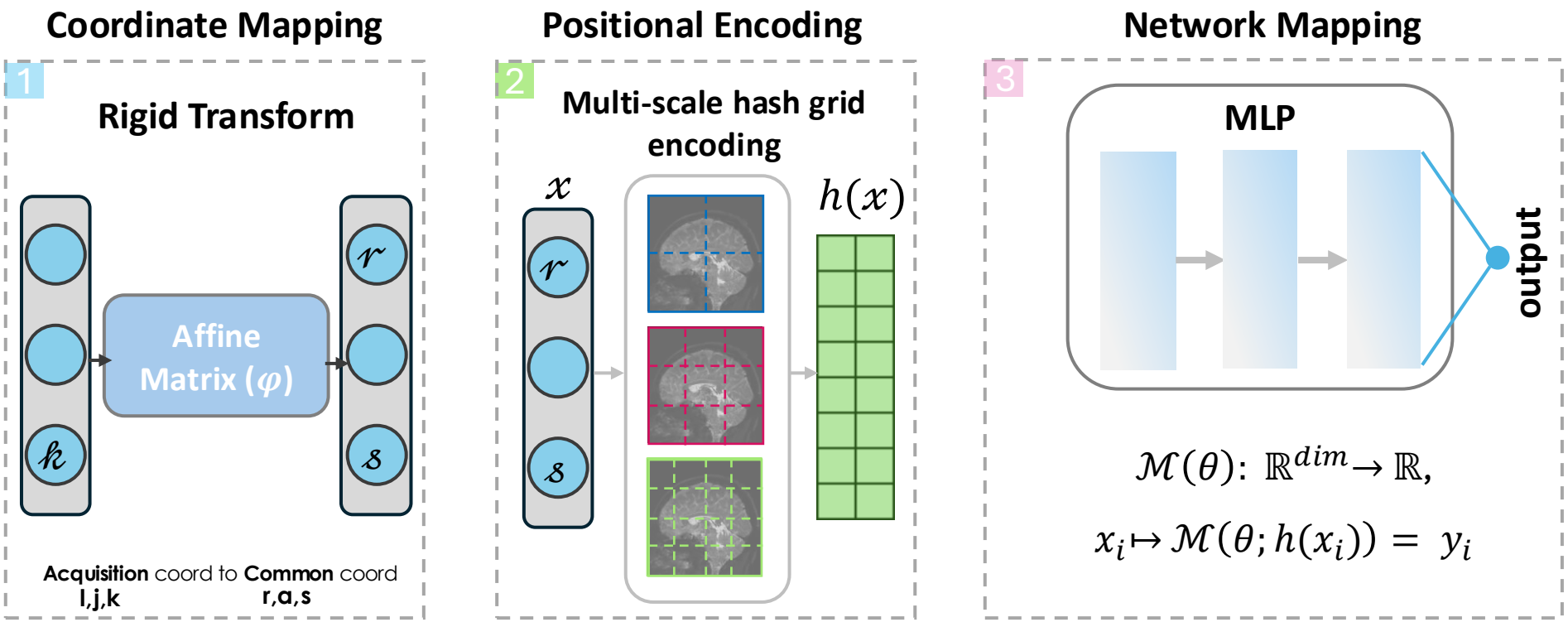
Enables: Cortical layer–specific diffusion, fine white matter pathways, …

2. **Challenge : SNR** drops rapidly with voxel size
3. **Idea:** acquire **high in-plane resolution, u**se **thick slices** to boost SNR, rotate the FOV across scans
4. Classic super-resolution reconstruction (**SRR**) $\hat{x} = argmin\ \|Ax - y\|_2^2 + \boldsymbol{\lambda R(x)}$
5. SRR with Implicit Neural Representation (**INR**), represents the image as a continuous function





ROVER-dMRI framework



- Training used **mini-batch SGD** (batch size = 100k) with **data-fidelity** + **total-variation** losses.

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \left\| L R_i^v - \hat{y}_i^{\mathrm{avg}} \right\|_2^2$$

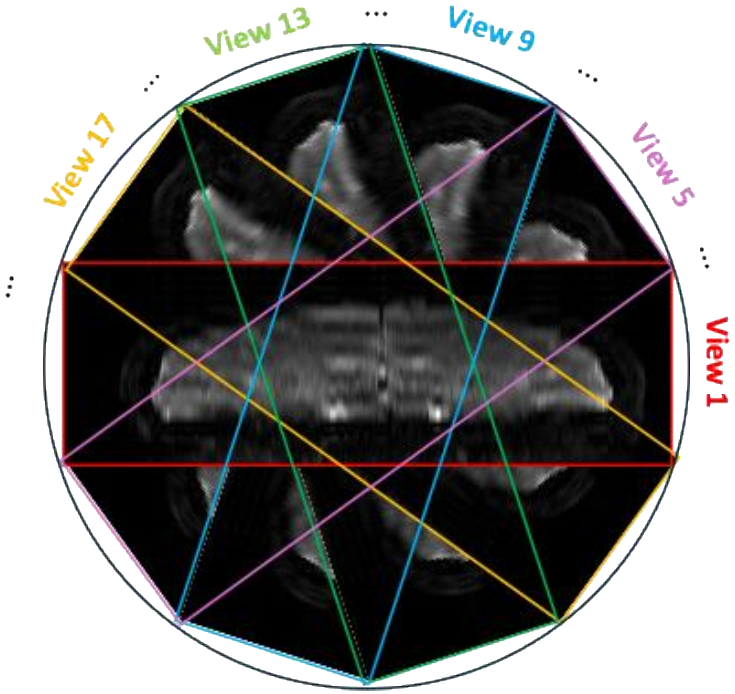
$$\mathcal{L}_{\mathrm{TV}} = \frac{1}{M} \sum_{j=1}^{M} \left\| \nabla \hat{y}_j \right\|_1 ,$$

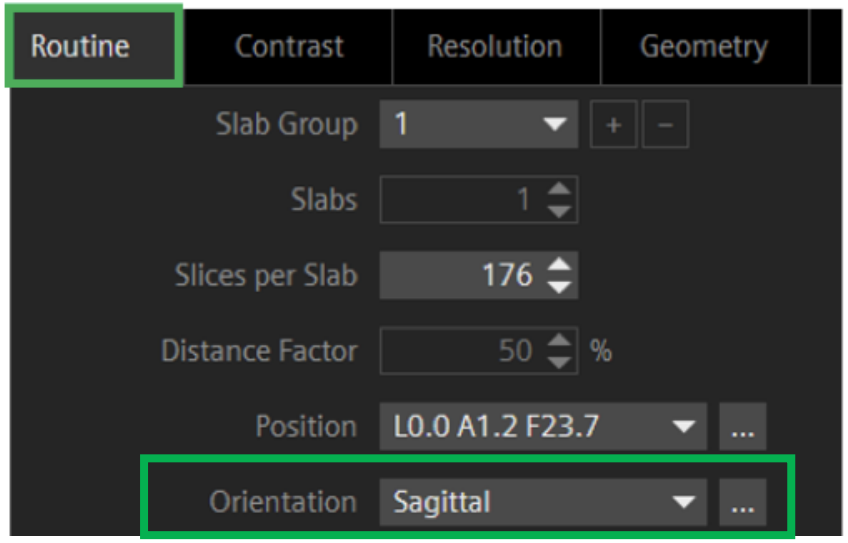
- x_i : spatial coordinate
- y_i : signal value at x_i
- M : neural network with parameters θ
- dim: spatial domain dimension

Müller, T. et al., 2022.
Hendriks, et al., 2025.
Lyu, et al., 2025.

ROVER-dMRI image acquisitions – rotating views

- **Rotate manually on the console (vendor-provided sequence)**





View #	Orientation
1	T 0
2	T-S 12
3	T-S 24
4	T-S 36
5	S-T 42
6	S-T 30
7	S-T 18

T : Transversal
S : Sagittal

ROVER-dMRI image acquisitions – rotating views with Pulseq

- **Rotate automatically in the Pulseq sequence (Pulseq V1.4.2)**

Rotation components

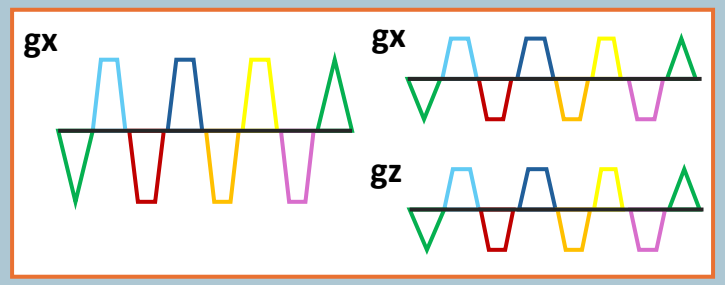
angle (φ)

axis (y)

Rotation matrix

$$\begin{bmatrix} \cos(\varphi) & 0 & \sin(\varphi) \\ 0 & 1 & 0 \\ -\sin(\varphi) & 0 & \cos(\varphi) \end{bmatrix}$$

Apply rotation to gradients



Transversal view Transverse-Sagittal 45°

mr.rotate/mr.rotate3D

```
%% sequence rotation
gx_comp1 = mr.rotate('y', alpha_rad, mr.scaleGrad(gx,dpg), mr.scaleGrad(gy_blipup, pe_enable*scale_gy_blipup),adc);
gx_comp2 = mr.rotate('y', alpha_rad, mr.scaleGrad(gx,dpg),mr.scaleGrad(gy_blipdown, pe_enable*scale_gy_blipdown),adc);
gx_comp3 = mr.rotate('y', alpha_rad, mr.scaleGrad(gx,dpg),mr.scaleGrad(gy_blipdownup, pe_enable*scale_gy_blipdownup),adc);

gxPre_post_comp = mr.rotate('y', alpha_rad, mr.scaleGrad(gxPre_post,dpg), mr.scaleGrad(gyPre_post,pe_enable*scale_gyPre_post));
gxPre_comp = mr.rotate('y', alpha_rad, mr.scaleGrad(gxPre,dpg),mr.scaleGrad(gyPre,pe_enable*scale_gyPre));

gz_comp = mr.rotate('y', alpha_rad, gz, rf);
gzReph_comp = mr.rotate('y', alpha_rad, gzReph);
gz180_comp = mr.rotate('y', alpha_rad, gz180, rf180);
```

ROVER-dMRI image acquisitions – rotating views with Pulseq

https://github.com/pulseq/pulseq

- **Rotate automatically in the Pulseq sequence (Pulseq V1.5.1)**

v1.5.1 Latest

m-a-x-i-m-z released this Aug 29, 2025 · 59 commits to master since this release v1.5.1 84ea726

A minor release with some important improvements and bug fixes. New features:

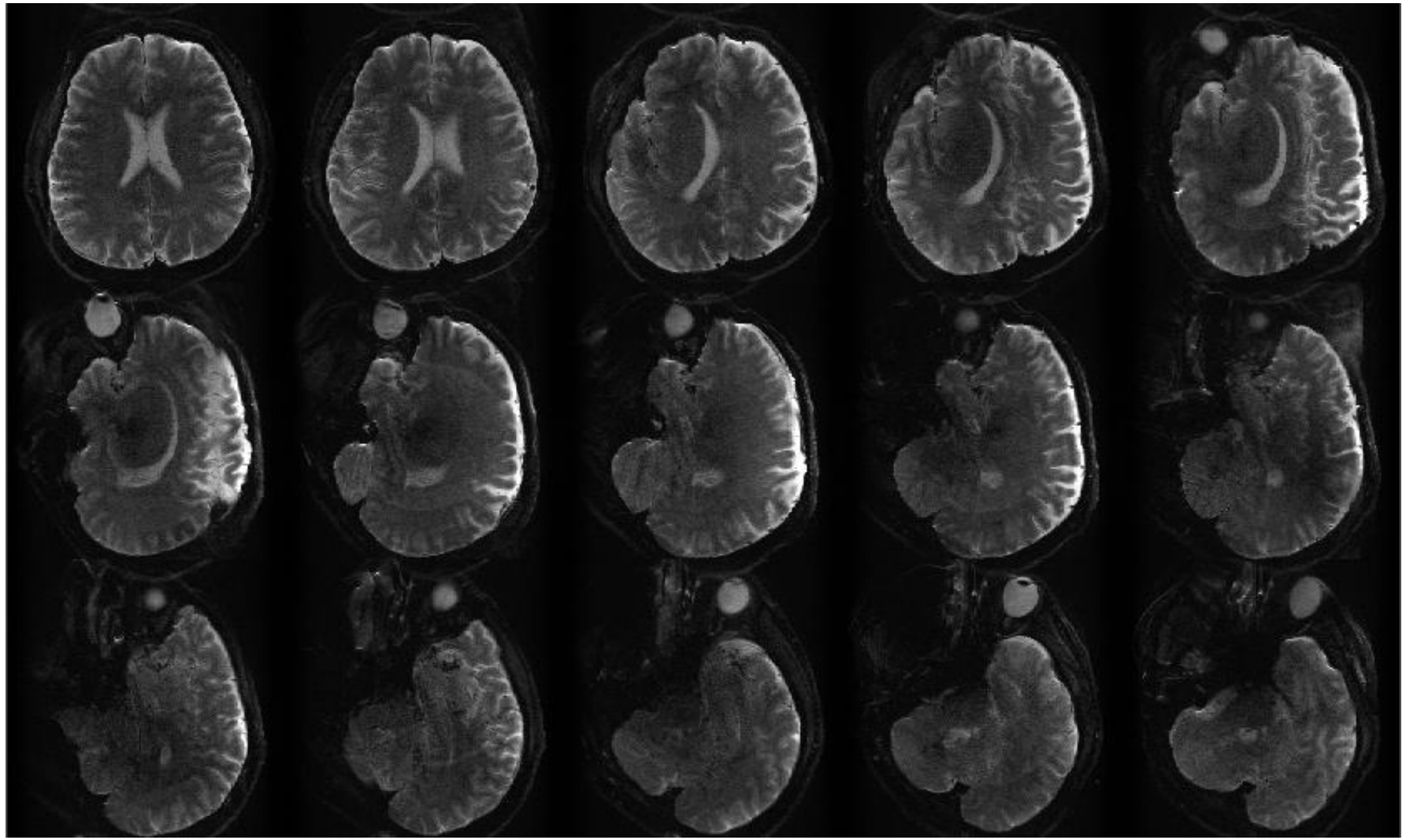
- Rotation extension
- RF shimming extension
- Plotting function shows triggers and digital output pulses
- Demo code for MR0 and Koma.jl signal simulations

```
for iview=[1:1]

    alpha_rad=rotation_angles(iview);
    rot_mat = [cos(alpha_rad), 0, sin(alpha_rad);0, 1, 0;-sin(alpha_rad), 0, cos(alpha_rad)];
    rot = mr.makeRotation(rot_mat);  % rotation event
                                ...
    seq.addBlock(gz,rf,rot,mr.makeLabel('SET', 'TRID', iview_fix));
    seq.addBlock(gzReph,rot);

                                ...
```

ROVER-dMRI – 0.4 mm iso with Pulseq on MAGNUS

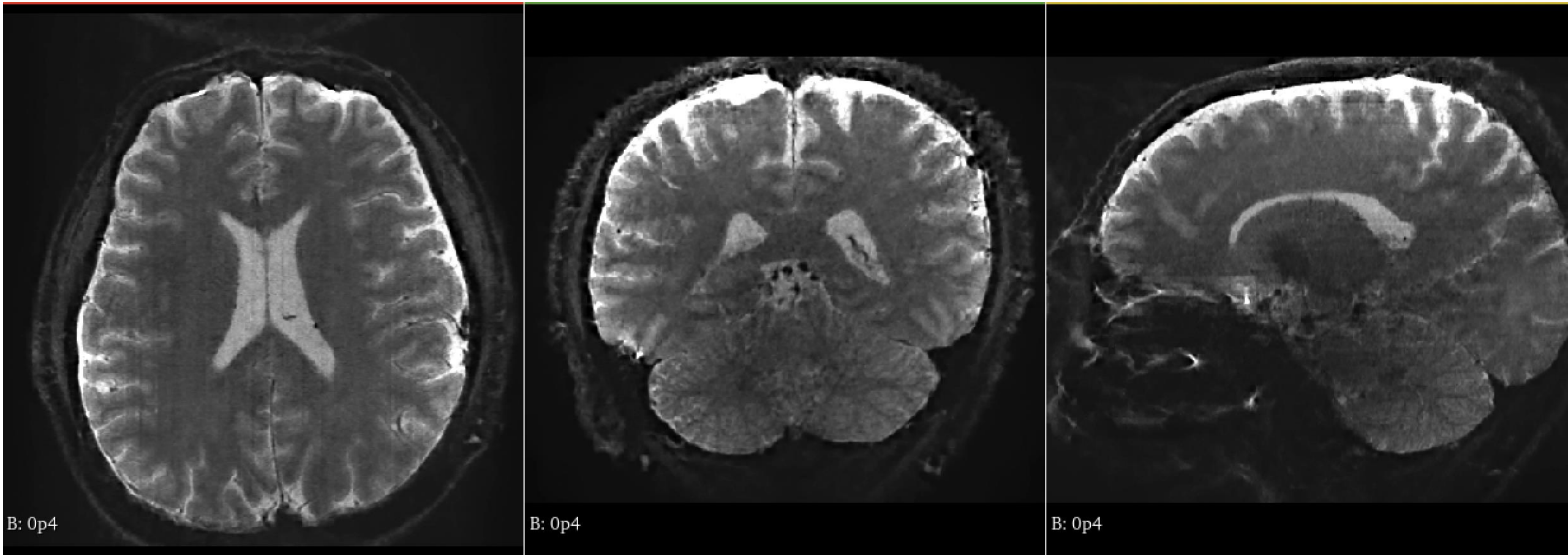
Low-resolution images

15 rotations, 0.4×0.4×6 mm^3, R=3



ROVER-dMRI – 0.4 mm iso with Pulseq on MAGNUS

Super-resolution images



0.4 mm iso

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Founding Member, Mass General Brigham

Psychiatry Neuroimaging Laboratory

HARVARD MEDICAL SCHOOL

Thank you for your attention!

qliu30@mgh.harvard.edu

Collaborators

Yunqi Wang

Nan Wang

Yiang Pan

Hwihun Jeong

Ante Zhu

Kawin Setsompop

Carl-Fredrik Westin

Berkin Bilgic

Borjan Gagoski

William Grissom

Lipeng Ning

Jon-Fredrik Nielsen

Maxim Zaitsev

Yogesh Rathi

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